

4.1 INTRODUCTION

The cells of an organism have a living material, protoplasm, which is sensitive, due to its sensitive nature it possess special property called irritability. It means it irritates when there is a change in its environment. Due to this irritability the living organism take some action to reduce its irritation these actions called movement. The change in environment takes place due to some factors these factors are called stimulus (stimuli: p). so we can say that movement is a response shown by living organisms toward stimuli. All living organism have property of locomotion due to nutrients, shelter and reproduction mainly plants are sessile while animals are usually motile therefore they respond to stimuli in different ways. Plant responds to stimuli by adjusting the rate of growth and osmotic condition. While respond to these stimuli by changing their position either towards or away from stimuli.

The movement can also occur at cellular level, like cyclosis in cell or the movement of chromosomes towards their respective poles during cell division etc.

On the basis of stimuli there are two types of movement

Autonomic or spontaneous movement

It is due to internal stimuli
i.e. cramps due to involuntarily release of Ca^{++} ions.

Paratonic or induced movement

It is due to external stimuli
i.e. reflex action.

Types of movement on the basis of responses.

Locomotory Movement or Taxis or Tactic Movement

- Types of movement where organism change its place either towards or away from stimulus

Trophic Movement

- Type of growth movement, organism move toward or away by growing their organs.
- Usually found in plants, fungi or in bacterial colony.

Nastic Movement

- Type of movement where change in osmotic water occur due to stimuli.



- | | | |
|--|---|--|
| <ul style="list-style-type: none"> • Usually found in animals, bacteria and protozoa. | <ul style="list-style-type: none"> • Growth of root towards water and mineral or growth of stem towards light. | <ul style="list-style-type: none"> • Purely found in plants like touch me not plant, which close leaves when touched. |
|--|---|--|

Difference between movement and locomotion

It is a misconception that locomotion and movement are the same phenomena. Movement is a broad term where organism response to stimuli in any way whereas locomotion is one of the type of movement where organism change its place takes place either towards stimuli or away from stimuli. If an organism show response towards stimuli but does not change its place it may be any type of movement but not locomotion.

4.2 SKELETON AND ITS TYPES

For locomotion animal body want support to change its place in a balanced and well-coordinated manner. For support animal have skeletal system. Skeleton are the frame work which gives shape to any structure. The skeleton performs three main functions

- Provide shape to organs
- Provide support to organs during movement
- Provide protection to soft, vital organs.

Types of skeleton

There are three main types of skeletons.

Hydrostatic Skeleton Hydro= water, static = to stay	Exoskeleton	Endoskeleton
• Skeleton made up of fluid. • Found in soft bodied animal. • Simplest type of skeleton.	• Skeleton deposit outside the body or organ. • Found in Arthropod, Mollusca and higher animals. • It provide support and protection.	• Skeleton develop inside the body. • Found in high animals. • Provide shape, support and protection.

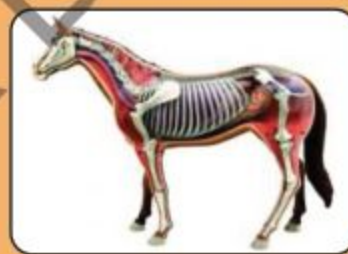
- Help in extension or withdrawal of body or its organs.
- In jelly fish, help in propulsion by water.
- In earthworm, coelomic fluid work as hydrostatic skeleton.



- Nonliving in nature, in arthropods made up of chitin, in Mollusca made up of CaCO_3 .
- In high animals made up of proteins.



- Living in nature made up of cells.
- Two types of endoskeleton cartilage made up of chondrocyte bones made up of osteocytes.



Human Skeleton

The skeletal system of human is basically made up of two types of skeletons

(i) **Cartilage**

(ii) **Bones**

Both are rigid, cellular structure and type of endoskeleton.

Cartilage

- Type of skeleton which is flexible
- It is made up of cells called chondrocytes embedded in a matrix of protein called collagen.
- It is much softer than bones as well as flexible.

Bones

- Type of skeleton which is harder
- It is made up of cell called osteocytes embedded in a matrix of protein called collagen.
- It is harder due to the deposition of calcium phosphate, process is called calcification.

- Calcification does not takes place.
- It covers ends of the bones and joints.
- No blood vessel penetrate into cartilage.

- Calcification takes place.
- It support the organ consists of an outer shell of compact bone.
- Blood vessels can penetrate into bone especially in spongy bones.

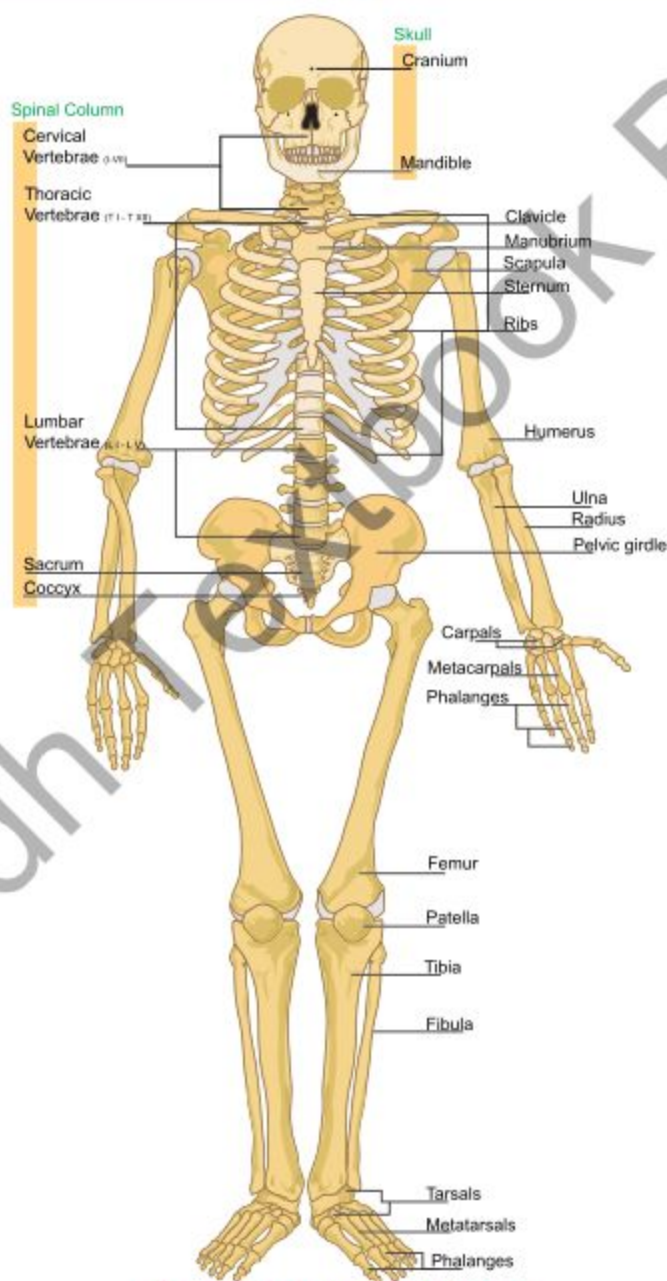


Fig. 4.1 Human Skeleton

4.2.1 Skeleton Provide Support and Movement

Skeleton provides support to the body, especially limbs. i.e. Hind Limbs and pelvic girdle provide support to the human body to be a bipedal animal, with these limbs the complete vertebrae of vertebral column provide support to stand straight. Cartilage of external pinna and nose support the organs to make their proper shape. Cartilages also support larynx, trachea and bronchi of respiratory system. It forms tough pads which acts as shock absorbers found in knee joints and also form intervertebral discs between vertebrae of backbone.



Fig. 4.2 Bone Forming Cells

4.2.2 Skeletal System is a Dynamic System

The dynamic property of skeletal system means that it is made up of living tissues, it is capable of quick growth. It can adapt to stress and can repair itself after damage (injury). 5% to 10% of our bones dissolve away annually and are replaced by a new one this process is called **remodeling** while in the growing age our bones enlarge with the growth of the body. The remodeling system allows a skeleton to alter the shape and size of skeleton in response to demands. e.g the bones carry heavy load or subjected to extreme stress become thicker to provide more strength and support. Normal stresses are major factors in maintaining bone strength.

There are three types of cells associated with bones i.e. bones forming cells (**osteoblast**), mature bone cell (**osteocyte**) and bone dissolving cells (**osteoclast**).

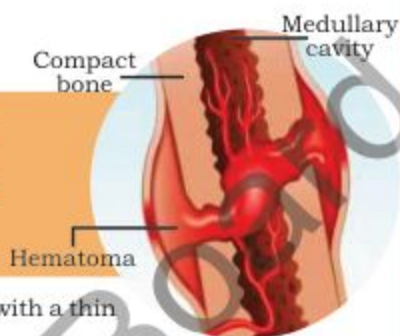
Bone remodeling is the result of coordinated activity of osteoclast and osteoblast. This coordination can be seen clearly at the time of repair of broken bone.

4.2.3 Model of Bone Healing Occurs During 6 Weeks.

A bone is broken. Let's see the sequence of events occur during 6 weeks of bone healing.

Step 1

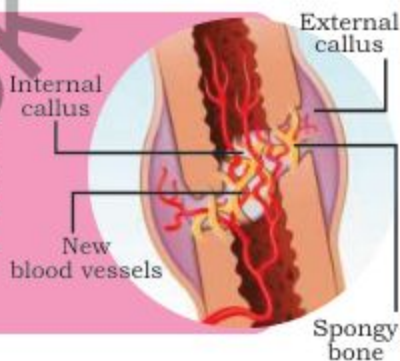
Blood from ruptured blood vessels form a large clot around the fracture. Phagocytic cells and osteoclast in the blood ingest, dissolved the Cellular debris and bone fragments.



*Bones are normally covered with a thin layer of connective tissues i.e. **periosteum**, rich in capillaries, osteoblast and osteoblast forming cells.

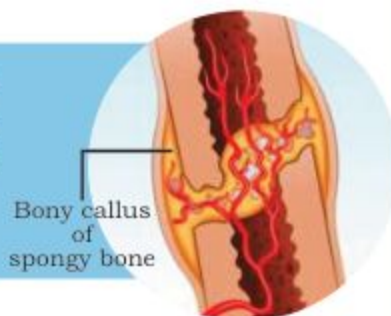
Step 2

A fracture ruptures the periosteum and stimulates the production and release of numerous osteoblasts, Which secrete a porous mass of bone and cartilage called callus, around break. Callus replaces the original blood clot and holds the end of the bones together, the remodeling process re-form the original shape of the bone.



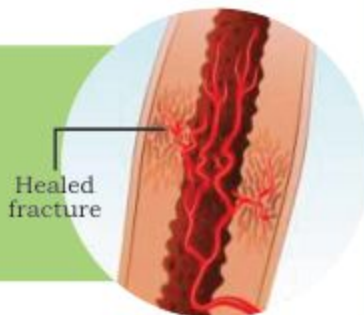
Step 3

Osteoclast, osteoblast and capillaries invade the callus. Blood nourished the cells, osteoclast breakdown the cartilage while osteoblast replace it with bone.



Step 4

The healing of bone completed



4.3 Human Skeleton

In humans 206 bones are present which can be categorized into two groups.

- (a) The axial skeleton
- (b) The appendicular skeleton

The **axial skeleton** forms the main axis of the human body, includes the bone of the head (skull 22 bones), vertebral column (26 vertebrae) and rib cage (ribs 12 pairs and sternum 1 bone).

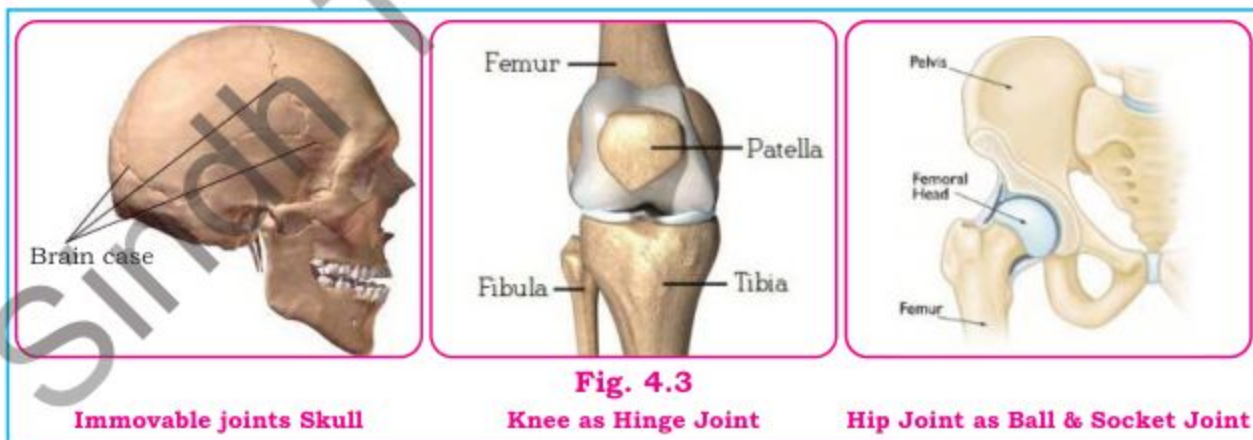
The **appendicular Skeleton** forms the appendages (Limbs) and their attachment to the axial skeleton includes pectoral (shoulder) and pelvis (hip) girdles.

Pectoral girdle consists of two bones i.e. scapula and clavicle. The forelimb consists of humerus, radius and ulna, carpal (8), metacarpal (5), and phalanges (14).

Pelvic girdle consists of three bones i.e. ilium, ischium and pubis. Hind limb consists of femur, patella, tibia, fibula, tarsals (7), metatarsals (5) and phalanges (14).

4.4 JOINTS

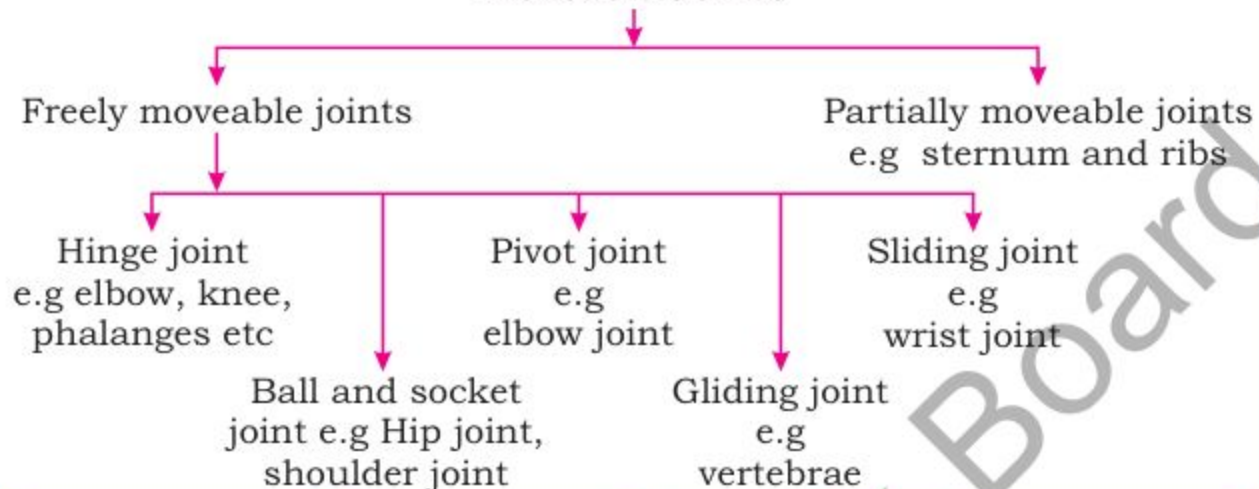
The junction of two bones is called a joint. There are two types of joints. Immovable or fixed joint. Joint where bones are fixed like puzzle pieces and do not allow to move, like 8 bones of skull or 3 bones of pelvic girdles.



Moveable joints where bones are allowed to move freely or partially. So there are two types of moveable joints i.e. freely moveable and partially moveable.



Moveable Joints



Hinge Joints (e.g knee joint) allow movement in one plane only whereas ball and socket (shoulder joint) allow movement in many plans such as forward, backward and sideways. Both hinge joints and ball socket joints have the same basic structure.

4.4.1 Ligament and Tendon: Their Role in Movement

The band of fibrous connective tissues by which bones are joined to one another at joints called **ligament**. It works as strong firmly attached ropes, it stabilizes the joint r hold the ends of two bones together. The strong onnective tissue in the gaments protects these tructures and prevents them om bending twisting or earing.

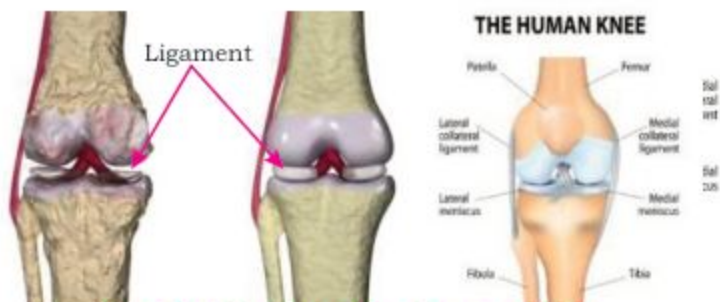


Fig. 4.5 Ligaments of knee joint

Skeletal muscles are attached to bones on either side of the joint by bands of tough, fibrous connective tissues called **tendons**. They are tougher and less elastic than ligaments. Tendon transfers the mechanical force of muscle contraction to the bones. It is strongly connected to muscle fiber at one end and to components of the bones at its other end. They are very strong, highly tensile.



Fig. 4.6
Bone tendon of skeletal muscles

4.4.2 Location and Movement of Hinge Joint

Hinge joints move back and forth like the hinge on a door and allow movements in one plane only. The knee and elbow are hinge joints.

4.4.3 Location and Movement of Ball and Socket Joints

This joint allows movement in all directions. The ball of humerus and femur fit into the socket of pectoral and pelvic girdle respectively. The hip and shoulder joints are ball and socket joints.

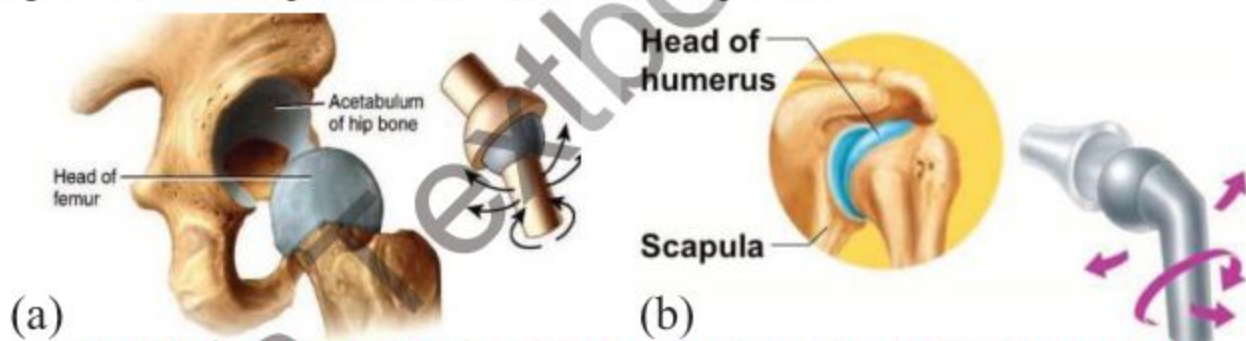


Fig. 4.7 Movement of Ball and socket of (a) Hip joint (b) Shoulder Joint

4.5 MUSCLES

Movements of arms and legs are brought about by the contraction of muscles attached to bones. Muscles are connective tissues consist of fibrous cells. These tissues have a tendency to contract and relax. The vertebrate possess three kind of muscles.

- i. Skeletal muscles
- ii. Cardiac muscles
- iii. Smooth muscles

i. Skeletal muscles

The muscles which are attached to the skeleton called skeletal muscles. They are associated with the movement of bones. These muscles are voluntary in nature. They are also called striped or stride muscles because they have alternate thick and thin means dark and light bands.

ii. Cardiac muscles

These are the muscles which build the walls of heart. They are also striated muscles but unlike skeletal muscles they are branched in nature and arranging mesh work. They are involuntary in nature, work under the control of SAN (Sino auricular node).

iii. Smooth muscles

Smooth muscles are the earliest form of muscles from evolutionary point of view. They are made up of long and spindle shaped cells, each cell contain single nucleus. They have no striations or stripes. They are involuntary in nature. These muscles are found in blood vessels, digestive tract and many other internal organs.

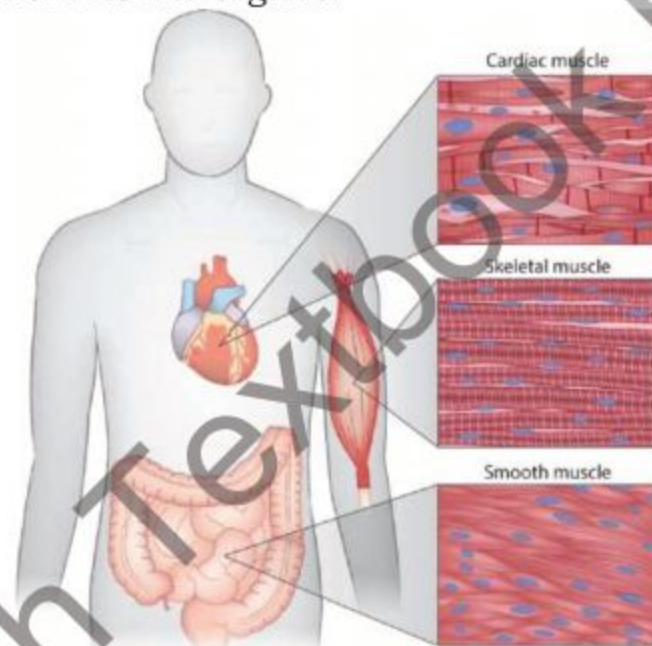


Fig. 4.8 Muscles

Support and locomotion is mainly due to skeletal system and skeletal muscles system. The contraction and relaxation of skeletal muscles allows the bones to move. The muscles which are attached with bones are called **skeletal muscles**. These muscles only exert a pulling force. When skeletal muscles relax, they are stretched by the contraction of another muscles. It shows that muscles are always found in pairs. The one muscle of this pair when contracted other muscles will relax. This type of working of two muscles against each other called **antagonism**. The pair of skeletal muscles which work against each other called an **antagonistic pair**. The action of **biceps** and **triceps** muscles of arm is a good example of an antagonistic pair.

The bicep muscle is a large muscle that lies on the front of the upper arm between the shoulder and elbow. It has two heads or origin. The tricep is also a large muscle on the back of arm. It is three headed or origin.

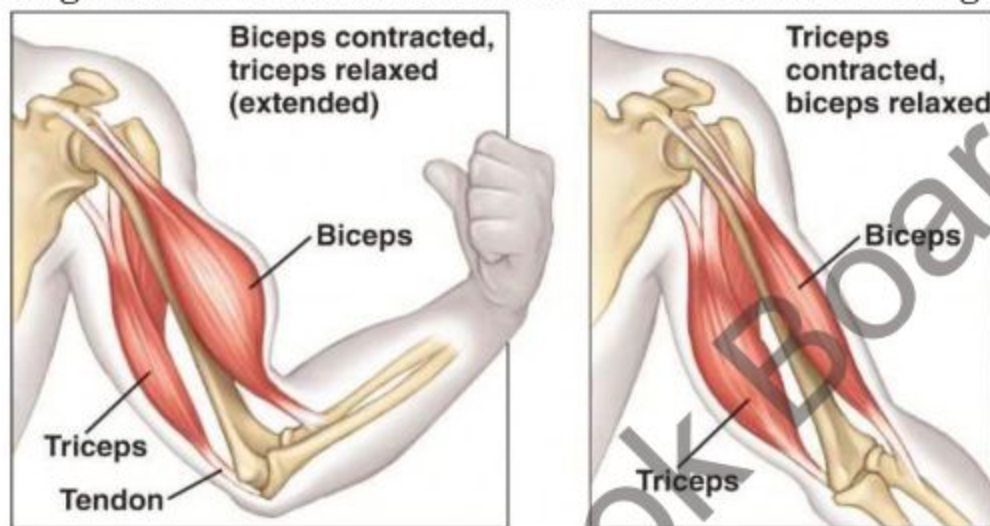


Fig. 4.9 Bicep and Tricep muscle work as Antagonistic muscles of arm

When the biceps muscle contracts it pulls upon the radius bone of lower arm, which bend the arm at elbow, this bending process is called **flexion**. On the other side when triceps muscle contracts it pull on the ulna which straightens or extends the arm. The straighten process is called **extension**. It shows that when the biceps contracts the triceps relaxes and vice versa. It means that in arm the biceps is the **flexor** muscle and **tricep** is the extensor muscle.

4.6 DISORDER OF SKELETON SYSTEM

4.6.1 Effect of Calcium Deficiency on Bone

The hardening of bones occurs due to deposition of calcium phosphate, this process is called calcification. If Ca^{++} deficiency occurs in body or blood, ultimately it occurs in bones. This deficiency of Ca^{++} in bone is called hypocalcemia known as calcium deficiency disease. A long term calcium deficiency can lead to dental weakness, osteoporosis, in childhood rickets.

Osteoporosis is a disorder related to the aging process. In this condition the bones become porous or more spongy, thinner and weaker so that they become fragile, in this condition a slight injury break the bone. Osteoporosis is more common in female than male.

Rickets

Rickets is the softening and weakening of bones in children, usually because of an extreme and prolonged vitamin D deficiency. Vitamin D

plays vital role in the absorption of Ca^{++} , deficiency of vitamin D leads to deficiency of Ca^{++} in bone.

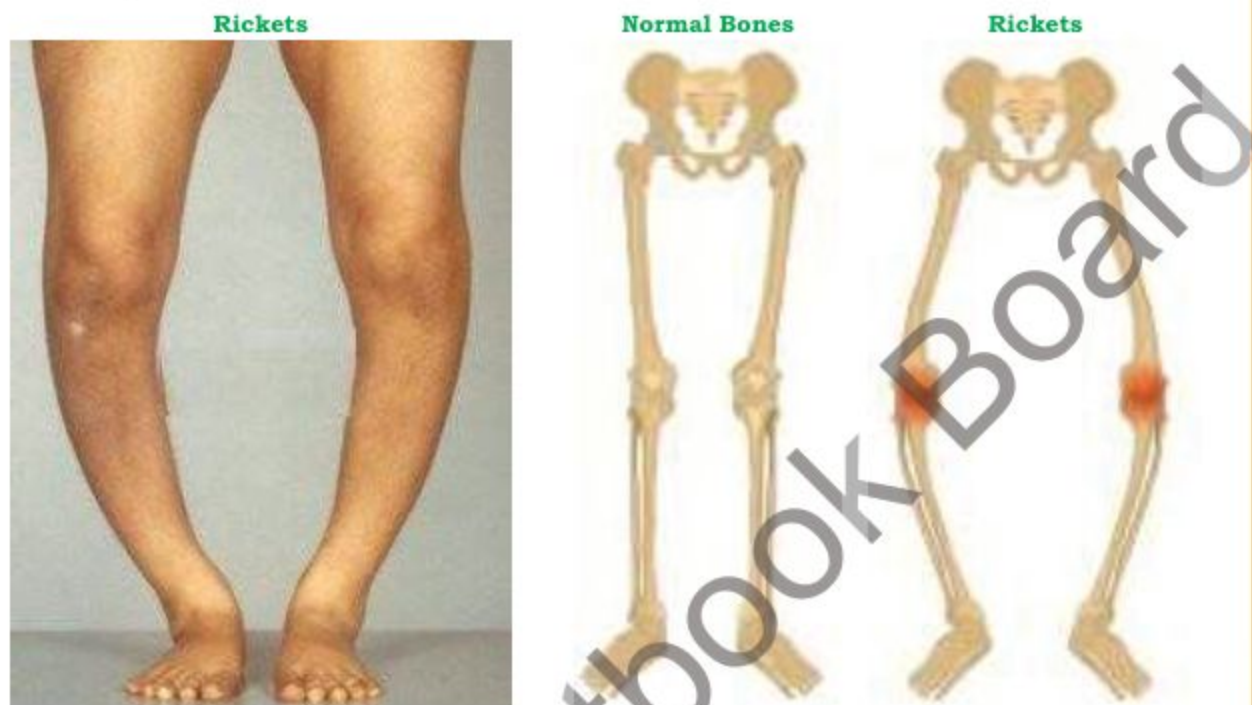


Fig. 4.10 Calcium deficiency develop rickets in children

4.6.2 Arthritis

It is the painful condition of joints due to swelling and tenderness of joint. The main symptom of arthritis are

- Joint pain
- Tenderness and stiffness
- Inflammation in and around the joint
- Restricted movement of joint
- Warm red skin over the affected joint
- Trouble dressing, combing, gripping, bending over squatty or climbing stair.



Fig. 4.11 Arthritis in knee as inflammation of joint

Common types of Arthritis

Osteoarthritis	It is a “wear and tear”, causes ; overuse of joints, age, joint injury, obesity. Joints that bear weight, like knees, hips, feet and spine.
Rheumatoid Arthritis	It is due to immune system i.e. autoimmune disorder. In this disorder immune system attacks on joints.
Psoriatic Arthritis	In this disorder joints and skin affects.

Causes of Arthritis

The following factors are the causes of arthritis.

- Injury
- Abnormal metabolism
- Infection
- Auto immune system
- Genetic makeup
- Reduction in the normal amount of cartilage or synovial fluid

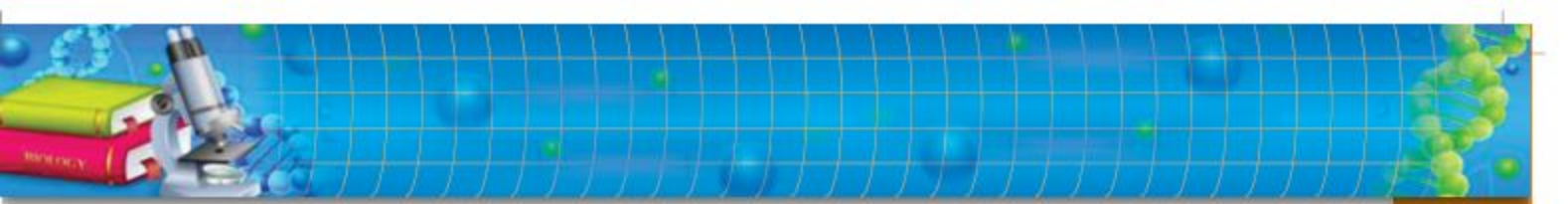
Role of obesity in arthritis

Obesity puts stress on joints, especially on knee, causing pain and develop worse condition in arthritis damage. The rheumatologist states that just 10 pound overweight increases force of 20 to 40 pound on knees at each step during walk.

The excessive fats tissues release high levels of cytokines proteins that can cause inflammation throughout body. The same protein produced by joints during rheumatoid arthritis. It makes existing joint inflammation more worse.

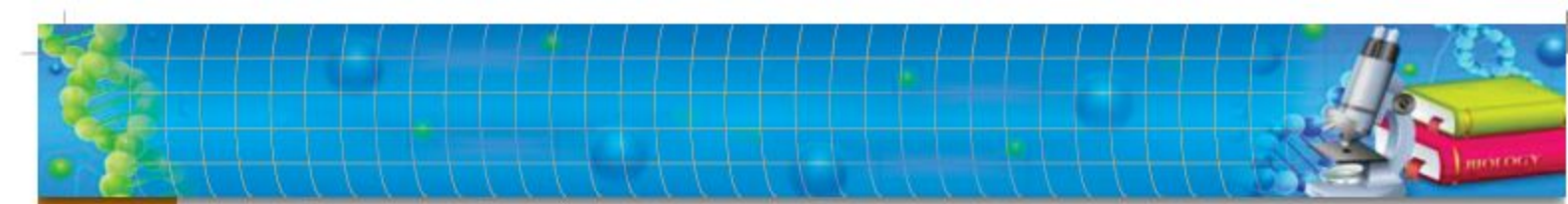


Fig. 4.12
Obesity as one of the cause of arthritis



SUMMARY

- ◆ Movement is the action taken by living organism to reduce its irritation.
- ◆ Locomotory movement require skeletal system, Muscular system
- ◆ Skeleton can be defined as frame work which gives shape to any structure.
- ◆ Skeleton provides support to the body, especially limbs. i.e. Hind Limbs and pelvic girdle.
- ◆ Cartilages also support larynx, trachea and bronchi of respiratory system.
- ◆ There are three types of cells associated with bones i.e. bones forming cells (**osteoblast**), mature bone cell (**osteocyte**) and bone dissolving cells (**osteoclast**).
- ◆ In humans 206 bones are present which can be categorized into two groups.
- ◆ The junction of two bones is called a **joint**.
- ◆ There are two types of joints. Immovable or fixed joint. Joint where bones are fixed like puzzle pieces.
- ◆ Hinge Joints (e.g knee joint) allow movement in one plane only whereas ball and socket allow movement in many plans.
- ◆ The band of fibrous connective tissues by which bones are joined to one another at joints called **ligament**.
- ◆ The strong connective tissue in the ligaments protects these structures and prevents them from bending twisting or tearing.
- ◆ Skeletal muscles are attached to bones on either side of the joint by bands of tough, fibrous connective tissues called **tendons**.
- ◆ This type of working of two muscles against each other called **antagonism**. This pair of skeletal muscles which work against each other called an **antagonistic pair**.
- ◆ This bending process is called **flexion**. Straighten process is called **extension**.
- ◆ A long term calcium deficiency can lead to dental weakness, osteoporosis, in childhood rickets.
- ◆ Osteoporosis is a disorder related to the aging process
- ◆ Rickets is the softening and weakening of bones in children, usually because of an extreme and prolonged vitamin D deficiency.
- ◆ Arthritis is the painful condition of joints due to swelling and tenderness of joint.
- ◆ Obesity puts stress on joints, especially on knee, causing pain and develop worse condition in arthritis damage.



EXERCISE

A. MULTIPLE CHOICE QUESTIONS

Choose the correct answer:

- i) Irritation caused by stimulus is the main cause of
(a) Tropism (b) Movement
(c) Locomotion (d) Arthritis
- ii) The frame work which gives shape to any structure called
(a) Architecture (b) Bone
(c) Cartilage (d) Skeleton
- iii) The cartilage are made up of cells called
(a) Osteoclast (b) Osteocytis
(c) Chondrocytes (d) Chaonocytes
- iv) The head of femur attached with
(a) Pelvic girdles (b) Pectoral girdle
(c) Scapula (d) Acetabulum of pelvic gridles
- v) The large muscles of arm is
(a) Bicep (b) Tricep
(c) Tetracep (d) Pentacep
- vi) The band of tough, fibrous, connective tissue which are attached to bone at joints called
(a) Ligament (b) Tendon
(c) Bicep (d) Tricep
- vii) Softening and weakening of bone in children due to deficiency of vitamin D called
(a) Osteoporesis (b) Osteoarthritis
(c) Rickets (d) Rheumatic fever
- viii) The muscle which is responsible to straighten the limb is
(a) Ligament (b) Skeleton muscle
(c) Flexor (d) Extensor
- ix) The pair of muscle where both work in opposite direction
(a) Antagonist (b) Cardiac
(c) Smooth (d) Abductor

B. SHORT QUESTIONS:

- i) What are the main functions of skeleton?
- ii) Distinguish between bone and cartilage.
- iii) Locate the cartilages in different organs.
- iv) Enlist the name and number of bones present in hind limb.
- v) Locate the immovable joints in the human body.
- vi) Draw neat and labeled diagram of hinge joint and ball and socket joint.
- vii) What is ricket? Give its causes.
- viii) Distinguish between ligament and tendon.
- ix) Locate hinge joint in the body of human
- x) What do you mean by bicep and tricep muscles?

C. EXTENSIVE RESPONSE QUESTIONS:

- i) Prove that the skeletal system is a dynamic system
- ii) Describe the flexion and extension in human arm
- iii) Explain disorders of bones and their causes.

Chapter 5

REPRODUCTION

Major Concept

In this Unit you will learn:

- Introduction
- Reproduction in Plants
 - ★ Asexual Reproduction in Plants
 - ★ Sexual Reproduction in Plants
- Asexual Reproduction in Animals
- Sexual Reproduction in Rabbit
 - ★ Male Reproductive System
 - ★ Female Reproductive System
- Population Planning
- Sexually Transmitted Diseases (AIDS)

